Dollar-Cost Averaging Versus Lump-Sum-Investing: Evidence from France

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Abstract

1. Introduction

Dollar-Cost Averaging (DCA) is an investment strategy whereby an investor allocates capital in equal amounts at regular intervals over a predefined investment horizon. This progressive investment mechanism is generally motivated by its ability to reduce timing risk, particularly the risk of committing a large amount of capital at market peaks. By construction, DCA leads investors to purchase more shares when prices are low and fewer when prices are high, thereby smoothing the average purchase cost over time (Dichtl and Drobetz, 2011). From a theoretical perspective, DCA can also be interpreted as a risk management device, acting as a partial hedge against short-term market volatility by spreading entry points across time (Milevsky and Posner, 2003).

In contrast, Lump-Sum Investing (LSI) consists of investing the entire available capital at a single point in time, thus providing immediate and full exposure to market risk. In upward-trending markets, this strategy is often expected to outperform DCA, as capital is deployed earlier and benefits more rapidly from market appreciation. Consequently, LSI may appear more attractive to investors with strong bullish expectations. Conversely, DCA tends to be preferred in environments characterized by heightened uncertainty or anticipated downturns (Grable and Chatterjee, 2015). However, such preferences implicitly rely on the ability to time market conditions, which remains notoriously difficult in practice.

This tension naturally raises a fundamental question for investors: which strategy should be preferred when entering financial markets under uncertainty? More specifically, beyond conditional preferences based on market expectations, it becomes essential to assess which approach delivers superior performance on average, and how sensitive this performance is to the timing of market entry.

The objective of this study is therefore to provide a comparative analysis of DCA and LSI strategies. In particular, we aim to contribute to the existing literature by offering new empirical evidence based on the French market. Additionally, this work seeks to enrich the debate by linking investment outcomes to behavioral considerations, thereby providing a more comprehensive understanding of how investors navigate the trade-off between risk and return when choosing between these two widely used investment strategies.

2. Methodology

2.1 Investment Framework and Assumptions

This study compares the performance of Dollar-Cost Averaging (DCA) and Lump-Sum Investing (LSI) on the French equity market. Both strategies are evaluated under a common set of assumptions, consistent with the existing literature (Leggio and Lien, 2001; Eriksson and Fransson, 2021).

An investor is assumed to have a fixed initial capital W_0 available at time $t = 0$. This capital represents a lump sum that could, in principle, be deployed immediately or phased into the market over a predefined horizon. The investor selects a single stock listed on Euronext Paris and commits to one of the two strategies without any subsequent tactical reallocation. No transaction costs are considered, and all dividends are assumed to be reinvested continuously, as reflected in the use of total return price series. Returns are expressed in nominal terms, with no adjustment for inflation. The uninvested cash under the DCA strategy is assumed to earn zero interest, consistent with the approach adopted by Eriksson and Fransson (2021) and representative of the current low-yield savings environment faced by retail investors in France. This assumption is conservative with respect to DCA, as it does not credit the cash reserve with any return during the phasing-in period, thereby introducing a mild bias in favour of LSI. This limitation is acknowledged and discussed in Section limitations.

2.2 Lump Sum Investing (LSI)

Under the Lump Sum Investing (LSI) strategy, the entire initial capital W_0 is fully deployed at the inception of the investment horizon t_0. At the effective start date t_0, shares are acquired at the observed market opening price $P_{\text{open}}(t_0)$. Fractional shares are fully preserved without rounding or residual cash drag, yielding an initial share count $N_{\text{LSI}}(t_0)$:

$$N_{\text{LSI}}(t_0) = \frac{W_0}{P_{\text{open}}(t_0)} \tag{1}$$

The dividend paid during the holding periods are fully reinvested using the dividend reinvestment plan (DRIP). Let $\mathcal{D} = \{t \in (t_0, t_F] \mid D_t > 0\}$ be the set of discrete dates within the investment window where a cash dividend per share D_t is distributed. On any dividend ex-date $t \in \mathcal{D}$, the accumulated cash payload is immediately converted into additional fractional shares at the daily market closing price $P_{\text{close}}(t)$. The share tracking engine updates iteratively:

$$N_{\text{LSI}}(t) = N_{\text{LSI}}(t^-) + \frac{N_{\text{LSI}}(t^-) \times D_t}{P_{\text{close}}(t)} \tag{2}$$

where $N_{\text{LSI}}(t^-)$ represents the share inventory held immediately prior to the dividend distribution at time t. The daily mark-to-market value of the portfolio, or equity curve $V_{\text{LSI}}(t)$, is calculated across the entire timeline by scaling the terminal share balance against the daily closing prices:

$$V_{\text{LSI}}(t) = N_{\text{LSI}}(t) \times P_{\text{close}}(t), \quad \forall t \in [t_0, t_F] \tag{3}$$

2.3 Dollar-Cost Averaging (DCA)

Under the Dollar-Cost Averaging (DCA) strategy, the initial capital W_0 is structured into systematic tranches to mitigate timing risk rather than being deployed as a single lump sum.

Let $\mathcal{T}_{\text{theory}}$ be the set of theoretical investment dates generated by the user-defined frequency freq $\in$ {D, W, M, Q, S, A}. Because theoretical dates can fall on weekends or market holidays, the execution engine maps each theoretical date to the first available active trading session on or after it. This yields an ordered set of N actual trading dates:

$$\mathcal{T} = \{t_1, t_2, \ldots, t_N\} \subseteq [t_0, t_F] \tag{4}$$

The total capital W_0 is divided into N equal allocations, determining a static cash tranche size W_{tranche}:

$$W_{\text{tranche}} = \frac{W_0}{N} \tag{5}$$

At each scheduled investment date $t_k \in \mathcal{T}$, the static tranche W_{tranche} is used to purchase additional asset fractions at the observed market opening price $P_{\text{open}}(t_k)$. Concurrently, the engine tracks discrete dividend distributions. Let $\mathcal{D} = \{t \in (t_0, t_F] \mid D_t > 0\}$ be the set of active dividend ex-dates.

The physical share inventory $N_{\text{DCA}}(t)$ updates sequentially for every chronological market date $t \in [t_0, t_F]$ according to the following cumulative matching logic:

$$N_{\text{DCA}}(t) = N_{\text{DCA}}(t^-) + \Delta N_{\text{tranche}}(t) + \Delta N_{\text{DRIP}}(t) \tag{6}$$

where $N_{\text{DCA}}(t^-)$ is the share balance from the prior trading day, and the incremental additions are defined as:

$$\Delta N_{\text{tranche}}(t) = \begin{cases} \dfrac{W_{\text{tranche}}}{P_{\text{open}}(t)} & \text{if } t \in \mathcal{T} \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

$$\Delta N_{\text{DRIP}}(t) = \begin{cases} \dfrac{N_{\text{DCA}}(t^-) \times D_t}{P_{\text{close}}(t)} & \text{if } t \in \mathcal{D} \text{ and } N_{\text{DCA}}(t^-) > 0 \\ 0 & \text{otherwise} \end{cases} \tag{8}$$

The daily mark-to-market equity curve $V_{\text{DCA}}(t)$ tracks the real-time valuation of the growing share baseline using daily closing prices across the entire observation window:

$$V_{\text{DCA}}(t) = N_{\text{DCA}}(t) \times P_{\text{close}}(t), \quad \forall t \in [t_0, t_F] \tag{9}$$

2.4 Rolling Window Simulation and performance metric

Rather than evaluating the strategies over a single fixed period, this study employs a multi-layered rolling window approach, consistent with the methodology of Leggio and Lien (2001) and Eriksson and Fransson (2021).

Formally, let t_{start} be the baseline historical anchor date. For a fixed investment horizon

$H = 5$ years and an execution step size $\Delta m =$ shift_months, the sequence of chronological window start dates $\{t_0^{(i)}\}_{i=1}^{M}$ and corresponding terminal boundaries $\{t_F^{(i)}\}_{i=1}^{M}$ are mapped sequentially via the index $i \in \{1, 2, \ldots, M\}$:

$$t_0^{(i)} = t_{\text{start}} + (i-1) \times \Delta m, \quad t_F^{(i)} = t_0^{(i)} + H \tag{10}$$

where M denotes the user-defined total number of independent rolling horizons. For each localized window i, the engine runs two isolated, path-dependent physical atomic simulations to resolve the underlying asset vectors and extract a clean cross-sectional matrix of three core performance structures:

Performance Measures

Extracted directly by evaluating the terminal mark-to-market capital yield against the localized initial deployment W_0:

$$\text{HPR}_{\text{strat}}^{(i)} = \frac{V_{\text{strat}}^{(i)}\left(t_F^{(i)}\right) - W_0}{W_0}, \quad \text{strat} \in \{\text{LSI, DCA}\} \tag{11}$$

The cross-strategy alpha differential for each rolling iteration is recorded directly into the temporal tracking series as:

$$\Delta\text{HPR}^{(i)} = \text{HPR}_{\text{LSI}}^{(i)} - \text{HPR}_{\text{DCA}}^{(i)} \tag{12}$$

Let $r_t^{(i)} = \ln\left(V_{\text{strat}}^{(i)}(t)/V_{\text{strat}}^{(i)}(t-1)\right)$ define the daily log return vector on active trading sessions inside window i. Given an annualized risk-free benchmark r_f, the daily excess asset return series is structured as $r_{e,t}^{(i)} = r_t^{(i)} - \frac{r_f}{252}$. The annualized Sharpe ($\mathcal{S}_h^{(i)}$) and Sortino ($\mathcal{S}_o^{(i)}$) profiles are modeled via standard sample variance and localized downside semi-variance frameworks:

$$\mathcal{S}_h^{(i)} = \sqrt{252} \times \frac{\mathbb{E}\left[r_{e,t}^{(i)}\right]}{\sigma\left(r_{e,t}^{(i)}\right)} \tag{13}$$

$$\mathcal{S}_o^{(i)} = \sqrt{252} \times \frac{\mathbb{E}\left[r_{e,t}^{(i)}\right]}{\sigma_{\text{down}}\left(r_{e,t}^{(i)}\right)}, \quad \text{where } \sigma_{\text{down}}\left(r_{e,t}^{(i)}\right) = \sqrt{\frac{1}{K-1}\sum_{t\in\text{neg}}\left(r_{e,t}^{(i)} - \mathbb{E}\left[r_{e,t}^{(i)}\right]\right)^2} \tag{14}$$

where $\text{neg} = \{t \mid r_{e,t}^{(i)} < 0\}$ represents the subset of trading days with negative excess performance, and $K = |\text{neg}|$.

To evaluate structural paths across business cycles (expansions, contractions, and market crises), each window is tagged with its baseline entry price level $P_{\text{open}}(t_0^{(i)})$. This acts as an exogenous conditioning variable used to map how the performance gap $\Delta\text{HPR}^{(i)}$ behaves across varying asset price regimes.

2.5 Statistical Framework and Hypothesis Testing

To rigorously assess whether the performance differences between the LSI and DCA strategies are statistically meaningful or merely artifacts of sample volatility, we implement a structured inferential framework. Given that our rolling window simulation evaluates both strategies over the exact same cross-sectional market paths within each window i, the resulting performance vectors are intrinsically paired.

2.5.1 *Normality and Specification Diagnostics*

Prior to conducting parametric inference, the distributional properties of the performance differentials $\Delta\text{HPR}^{(i)}$ are formally evaluated to verify the underlying assumptions of ordinary least squares and classical standard errors. We apply the Shapiro-Wilk test for normality, which computes the test statistic W:

$$W = \frac{\left(\sum_{i=1}^{M} a_i \Delta\text{HPR}^{(i)}\right)^2}{\sum_{i=1}^{M}\left(\Delta\text{HPR}^{(i)} - \overline{\Delta\text{HPR}}\right)^2} \tag{15}$$

where a_i represents the weights derived from the covariances of the order statistics of a standard normal distribution, and $\overline{\Delta\text{HPR}}$ is the sample mean differential. Under the null hypothesis H_0^{SW}, the differences are normally distributed. If H_0^{SW} is rejected at a significance level of $\alpha = 0.05$, non-parametric alternatives (such as the Wilcoxon Signed-Rank test) are reported to complement the parametric inference.

2.5.2 *Paired Student's t-Test*

Subject to satisfactory specification diagnostics, a two-sided paired Student's t-test is deployed to test the null hypothesis that the true mean performance differential between LSI and DCA is equal to zero, against the alternative hypothesis that their expected returns diverge:

$$\begin{cases} H_0 : \mu_{\Delta\text{HPR}} = 0 \\ H_1 : \mu_{\Delta\text{HPR}} \neq 0 \end{cases} \tag{16}$$

By accounting for the path-dependent covariance common to both strategies within each simulation window, the paired t-statistic eliminates the cross-sectional variance drag, optimizing statistical power. The test statistic t is formulated as:

$$t = \frac{\overline{\Delta\text{HPR}}}{s_\Delta / \sqrt{M}} \tag{17}$$

where s_Δ represents the sample standard deviation of the paired differences, defined as:

$$s_\Delta = \sqrt{\frac{1}{M-1}\sum_{i=1}^{M}\left(\Delta\text{HPR}^{(i)} - \overline{\Delta\text{HPR}}\right)^2} \tag{18}$$

The computed statistic follows a Student's t-distribution with $M - 1$ degrees of freedom. To determine the directionality of the outperformance under a stricter framework, we also compute one-sided upper tail p-values testing $H_0 : \mu_{\Delta\text{HPR}} \leq 0$ against $H_1 : \mu_{\Delta\text{HPR}} > 0$ to formally isolate the statistical significance of the LSI premium.

3. Results

3.1 Cross-Sectional Analysis on the Aggregate CAC 40 Index

Rolling window simulations over the period starting in 2010 indicate a structural dominance of the Lump-Sum Investing (LSI) strategy over Dollar-Cost Averaging (DCA). The only tactical horizons where DCA marginally outperforms LSI correspond to entry dates spanning from early 2010 to late 2011.

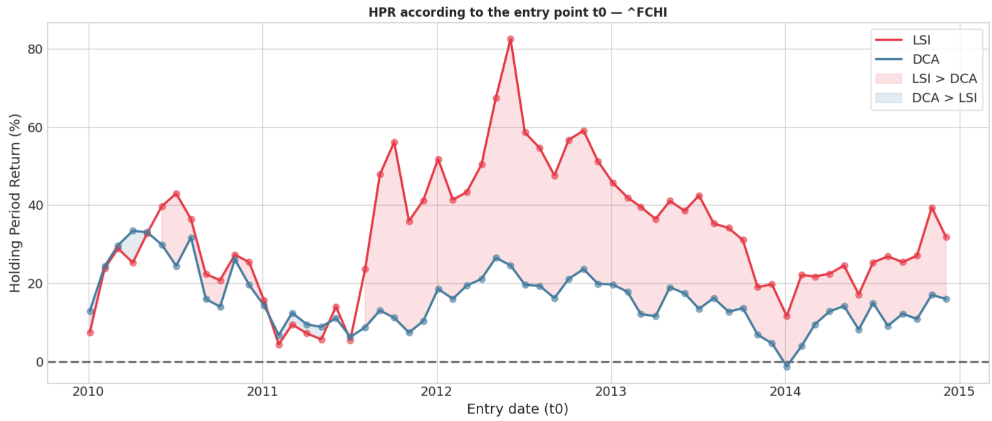

Figure 1. Temporal evolution of LSI vs. Monthly DCA HPR across rolling 5-year horizons (2010–2015).

This specific outperformance window is fundamentally linked to the macroeconomic regime of the French equity market at that time. Following the 2008 financial crisis and structural shocks from the Eurozone sovereign debt crisis (notably the Greek default risks), the CAC 40 experienced high volatility and a prolonged sideways-to-negative trend. In such non-directional or declining markets, DCA mechanically mitigates timing risk by acquiring cheaper fractions of shares, lowering the average cost basis. Conversely, as soon as the index enters a sustained secular bull regime post-2012, LSI captures the full upward beta from day one, exposing the cash drag inherent to DCA.

3.1.1 Sensitivity Analysis to DCA Infusion Frequency

Table 1 provides a comparative analysis of alternative DCA scheduling regimes (Monthly, Semi-Annual, and Annual) against the baseline LSI.

Table 1. CAC 40 Index Backtest Framework Summary ($\Delta t = 5$ Years, $M = 60$ Windows, $W_0 =$ 12,000).

DCA Frequency	Metric	LSI Mean	LSI Std	DCA Mean	DCA Std	LSI Wins (%)
Monthly	HPR (%)	33.11	16.46	19.63	7.93	76.67
(12×/year)	CAGR (%)	5.76	2.60	3.62	1.36	76.67
	W_{final} ()	15,973.62	1,975.59	14,355.45	951.68	76.67
Semi-Annual	HPR (%)	33.11	16.46	20.76	8.27	73.33
(2×/year)	CAGR (%)	5.76	2.60	3.81	1.41	73.33
	W_{final} ()	15,973.62	1,975.59	14,491.58	991.96	73.33
Annual	HPR (%)	33.11	16.46	20.89	7.42	73.33
(1×/year)	CAGR (%)	5.76	2.60	3.84	1.26	73.33
	W_{final} ()	15,973.62	1,975.59	14,506.40	890.70	73.33

The empirical evidence confirms that reducing the sampling frequency from monthly to semi-annual capitalizes on front-loading larger tranches of capital earlier in the window.

The Semi-Annual DCA exhibits a higher mean HPR (20.76%) compared to the Monthly scheme (19.63%). Interestingly, the transition from semi-annual to annual execution yields nearly identical, saturated returns (20.89%), showing a marginal further increase rather than a decay in performance. Across all configurations, LSI yields a systematic win rate ranging from 73.33% to 76.67%, exhibiting higher expected returns at the expense of increased variance ($\sigma_{\text{LSI}} \approx 16.46\%$ vs. $\sigma_{\text{DCA}} \approx 7.42\%$–$8.27\%$).

3.2 Cross-Sectional Granular Analysis across CAC 40 Individual Stocks

The cross-sectional analysis across individual constituents of the CAC 40 confirms that the outperformance of Lump-Sum Investing (LSI) observed at the aggregate index level remains highly robust when broken down by individual stock tickers.

Table 2. Cross-Sectional Sector Analysis: Lump-Sum Investing (LSI) vs. Semi-Annual DCA on CAC 40 Components (M = 60 Rolling Windows)

	Lump-Sum Investing (LSI)		**Semi-Annual DCA (2×/year)**		**LSI Win**
Company	**HPR (%)**	**Std**	**HPR (%)**	**Std**	**Rate (%)**
Capgemini	173	58	84	33	98
Michelin	90	40	44	19	100
Renault	113	78	48	48	100
Stellantis	241	141	115	53	87
Schneider Electric	48	29	26	13	90
Crédit Agricole	93	107	52	31	67
BNP Paribas	47	52	26	22	67
AXA	107	53	52	27	93
Société Générale	47	76	25	31	60
Air Liquide	57	17	31	12	100
Vinci	121	42	64	14	98
Engie	−2	16	5	10	32
Orange	70	36	43	19	63
TotalEnergies	50	18	28	8	95
Veolia	92	62	59	26	67
EssilorLuxottica	91	39	41	23	100
Sanofi	56	40	24	21	97
Eurofins Scientific	382	229	129	76	100
Airbus	195	59	85	23	100
Alstom	13	39	20	23	32
Thales	205	71	99	32	97
Safran	201	49	90	21	100
Legrand	106	38	49	19	100
Saint-Gobain	34	35	19	19	75
Christian Dior	151	58	83	36	100
Kering	161	83	97	60	100
LVMH	107	50	63	31	100
L'Oréal	108	23	53	14	100
Hermès	130	51	62	14	97
Accor	58	32	30	23	92
Danone	50	12	28	8	100
Carrefour	0	32	3	24	42
Pernod Ricard	75	20	41	12	100
Bouygues	75	52	44	21	68
Publicis Groupe	55	54	19	29	90
Teleperformance	344	120	153	26	100
Vivendi	170	34	81	18	100
Dassault Systèmes	163	27	78	16	100

Across nearly all companies evaluated, LSI delivers a substantially higher mean Holding Period Return (HPR) compared to the Semi-Annual DCA strategy, capturing a systematic premium from immediate market exposure. For instance, high-growth or volatile assets such as Vivendi, Teleperformance, and Eurofins Scientific display the most dramatic absolute performance gaps, where LSI leverages compound returns on early capital commitment.

Concurrently, the empirical evidence illustrates the classic risk-return trade-off: the higher expected returns of LSI are systematically accompanied by an elevated standard deviation (Std) across all assets. In contrast, the Semi-Annual DCA effectively acts as a volatility smoothing mechanism, significantly compressing the standard deviation of terminal outcomes.

Nevertheless, this variance reduction comes at a steep opportunity cost. LSI maintains a dominant win rate, reaching 100% for the vast majority of individual stocks. The rare structural exceptions where DCA demonstrates competitive performance or a lower LSI win rate—such as Engie and Alstom (LSI win rate of 32% each, the lowest in the sample), Société Générale (60%), Carrefour (42%), and Crédit Agricole and BNP Paribas (67% each)—coincide with assets that experienced prolonged sideways or secular downward trends over the historical simulation windows. In these specific configurations, the systematic phasing of DCA mechanically mitigated downside risk by capturing lower average purchase costs, confirming that DCA's relative value is strictly conditional on non-bullish underlying regimes.

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